THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 4410 cm³	Sphere $= \frac{4}{3}\pi\left(\frac{21}{2}\right)^3 = 4851$ cm³; cube $= 9261$ cm³; Wasted $= 4410$ cm³
2	**(c)** 23	$\frac{4325+25p}{77+p} = 49$ gives $(25-49)p = -552$; $p = 23$
3	**(d)** $\frac{5}{36}$	Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$
4	**(a)** 2425.5 cm³	Cylinder $= \pi r^2(2r) = 7276.5$ cm³; sphere $= \frac{4}{3}\pi r^3 = 4851$ cm³; Unoccupied $= 2425.5$ cm³
5	**(d)** 9	$238.4 = 230 + \frac{50-(20+x)}{25} \times 10$ gives $x = 9$; $x = 9$; then $y = 100 - 82 - x = 9$
6	**(d)** $\frac{1}{6}$	Favourable outcomes: (1,1), (1,2), (1,3), (2,1), (2,2), (3,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
7	**(b)** 15 cm	$H = 3h = 3 \times 5$; $H = 15$ cm
8	**(a)** 269.8	$\sum f_i = 100$ and $\sum f_i x_i = 26980$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{26980}{100} = 269.8$
9	**(b)** 45	$\frac{x}{15+x} = 3 \times \frac{15}{15+x}$ gives $x = 3 \times 15$; $x = 45$
10	**(c)** 14 cm	Cancel $\frac{1}{3}\pi r^2$: $h = 2r$; $h = 14$ cm
11	**(c)** 108	Median class 100–120: $l = 100$, $cf = 44$, $f = 15$, $h = 20$; Median $= l + \frac{\frac{N}{2}-cf}{f} \times h = 100 + \frac{50-44}{15} \times 20 = 108$
12	**(d)** 5	Cross-multiplying: $7(7+x) = 4(16+x)$; $x = 5$
13	**(a)** Both A and R are true and R is the correct explanation of A.	$2\pi r^2 + \pi r^2 = 3\pi r^2$; R gives the reason.
14	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{10+20}{2} = 15$; R is the definition used.
15	**(a)** Both A and R are true and R is the correct explanation of A.	No face of a die shows more than 6, so the event is impossible and its probability is 0.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(6)^3 = \pi(8)^2 x$ (1); $x = 4.5$ cm (1).	1+1
17	$\sum f_i = 83 + p$ and $\sum f_i x_i = 23150 + 250p$; $\frac{23150+250p}{83+p} = 274$ gives $(250-274)p = -408$ (1); $p = 17$ (1).	1+1
18	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $= 9$ (1); $P = \frac{9}{36} = \frac{1}{4}$ (1).	1+1
19	Volume of material = outer hemisphere − inner hemisphere; $\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(7.5^3 - 4.5^3)$ (1); Volume $= 693$ cm³ (1).	1+1
20	Highest frequency is 23, so the modal class is 20–30; $l = 20$, $f_1 = 23$, $f_0 = 4$, $f_2 = 4$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{19}{38} \times 10 = 25$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Volume of cylinder $= \pi \times 9^2 \times 32$ (1); Volume of a sphere $= \frac{4}{3}\pi(6)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 9$ (1).	1+1+1
22	$\sum f_i = 100$ and $\sum f_i x_i = 7530$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7530}{100} = 75.3$ (1); $l = 80$, $f_1 = 23$, $f_0 = 20$, $f_2 = 2$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{3}{24} \times 10 = 81.25$ (1).	1+1+1
23	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (3,6), (4,5), (5,4), (6,3); number $= 4$ (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1+1
24	Volume of sphere = volume of cone (1); $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(10)^2(5)$ gives $R^3 = 125$ (1); $R = 5$ cm (1).	1+1+1
	OR: Take $a = 75$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i = 100$ and $\sum f_i u_i = -70$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 75 + 10 \times \frac{-70}{100} = 68$ (1).	1+1+1
25	$\frac{9+x}{16+x} = \frac{2}{3}$ (1); Cross-multiplying: $3(9+x) = 2(16+x)$ (1); $x = 5$ (1).	1+1+1

	OR: Volume of cylinder $= n\times$ volume of one cone (1); $\pi(6)^2(20) = n \times \frac{1}{3}\pi(3)^2(8)$ (1); $n = 30$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $2\pi rh = 2 \times \frac{22}{7} \times 3.5 \times 7 = 154\ \text{mm}^2$ (1). (ii) Two hemispheres make a sphere: $4\pi r^2 = 4 \times \frac{22}{7} \times 12.25 = 154\ \text{mm}^2$ (1). (iii) $154 + 154 = 308\ \text{mm}^2$ (1). (iv) Length $= 7 + 3.5 + 3.5 = 14$ mm (1); volume of cylinder $= \frac{22}{7} \times 12.25 \times 7 = 269.5\ \text{mm}^3$ (1).	1+1+1+2
27	Total outcomes $= 36$ (2); Favourable outcomes: 4; 8; 15 (2); Probabilities: $\frac{1}{9}$, $\frac{2}{9}$, $\frac{5}{12}$ (1).	2+2+1
	OR: The cylinder has radius r and height $2r$ (2); Cylinder $= \pi r^2(2r) = 196465.5$ cm^3; sphere $= \frac{4}{3}\pi r^3 = 130977$ cm^3 (2); Unoccupied $= 65488.5$ cm^3 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 8 : 27	$2^3 : 3^3 = 8 : 27$; Ratio $= 8 : 27$
2	**(b)** 25	$20 + \frac{25-18}{14} \times 10 = 25$
3	**(d)** $\frac{19}{70}$	Favourable numbers: 19 outcomes; number $= 19$; $P = \frac{19}{70}$
4	**(d)** 30	$\pi(6)^2(10) = n \times \frac{1}{3}\pi(3)^2(4)$; $n = 30$
5	**(c)** 20	$\frac{14660+190p}{80+p} = 184.6$ gives $(190 - 184.6)p = 108$; $p = 20$
6	**(a)** $\frac{43}{50}$	Favourable outcomes = number of good pens; $P = \frac{215}{250} = \frac{43}{50}$
7	**(b)** 643.5 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(9^3 - 7.5^3)$; Volume $= 643.5$ cm^3
8	**(c)** 21	$25 = 20 + \frac{50-(20+x)}{18} \times 10$ gives $x = 21$; $x = 21$; then $y = 100 - 75 - x = 4$
9	**(c)** $\frac{1}{6}$	Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
10	**(c)** 35280 cm³	Sphere $= \frac{4}{3}\pi\left(\frac{42}{2}\right)^3 = 38808$ cm^3; cube $= 74088$ cm^3; Wasted $= 35280$ cm^3
11	**(b)** 71	$\sum f_i = 100$ and $\sum f_i x_i = 7100$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7100}{100} = 71$
12	**(c)** $\frac{2}{9}$	Favourable outcomes: (1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (6,4); number $= 8$; $P = \frac{8}{36} = \frac{2}{9}$
13	**(c)** A is true but R is false.	$\frac{1}{3}\pi r^2 h$ against $\pi r^2 h$: A is true; the cone volume is $\frac{1}{3}\pi r^2 h$, so R is false.
14	**(c)** A is true but R is false.	A is the correct rule; the class with the highest frequency is the *modal* class, so R is false.
15	**(c)** A is true but R is false.	Primes are 2, 3, 5, so $P = \frac{3}{6} = \frac{1}{2}$: A true; probability lies between 0 and 1, so R is false.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$P(\text{blue}) = \frac{x}{18+x}$ and $P(\text{red}) = \frac{18}{18+x}$; $\frac{x}{18+x} = 3 \times \frac{18}{18+x}$ gives $x = 3 \times 18$ (1); $x = 54$ (1).	1+1
17	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(12)^3 = \pi(2)^2 L$ (1); $L = 576$ cm (1).	1+1
18	Cumulative frequencies: 23, 27, 52, 58, 61, 80, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 240–260: $l = 240$, $cf = 27$, $f = 25$, $h = 20$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 240 + \frac{50-27}{25} \times 20 = 258.4$ (1).	1+1
19	After adding, red balls $= 15 + 6 = 21$ and total balls $= 30$; Favourable outcomes = number of red balls (1); $P = \frac{21}{30} = \frac{7}{10}$ (1).	1+1
20	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(9)^3 = \pi(12)^2 x$ (1); $x = 6.75$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Median class is 230–240 with $f = 20$, and the cumulative frequency before it is $cf = 24 + x$ (1); $239.5 = 230 + \frac{50-(24+x)}{20} \times 10$ gives $x = 7$ (1); $x = 7$; then $y = 100 - 80 - x = 13$ (1).	1+1+1
22	Total possible outcomes $= 84$ (1); Favourable numbers: 2; number $= 1$ (1); $P = \frac{1}{84}$ (1).	1+1+1
23	Volume of cylinder $= \pi \times 6^2 \times 72$ (1); Volume of a sphere $= \frac{4}{3}\pi(6)^3$; $n = \frac{3r^2 h}{4\rho^3}$ (1); $n = 9$ (1).	1+1+1
24	$\sum f_i = 100$ and $\sum f_i x_i = 10360$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{10360}{100} = 103.6$ (1); $l = 120$, $f_1 = 24$, $f_0 = 15$, $f_2 = 17$, $h = 20$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 120 + \frac{9}{16} \times 20 = 131.25$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6); number $= 9$ (1); $P = \frac{9}{36} = \frac{1}{4}$ (1).	1+1+1
25	Volume of sphere = volume of cone (1); $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(8)^2(4)$ gives $R^3 = 64$ (1); $R = 4$ cm (1).	1+1+1
	OR: Take $a = 270$, $h = 20$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i = 100$ and $\sum f_i u_i = -28$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 270 + 20 \times \frac{-28}{100} = 264.4$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $\frac{30+40}{2} = 35$ (1). (ii) 40-50, the class with the highest frequency 14 (1). (iii) Class marks 15, 25, 35, 45, 55; $\sum fx = 30 + 100 + 350 + 630 + 550 = 1660$; mean $= \frac{1660}{40} = 41.5$ min (1). (iv) Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h = 40 + \frac{14 - 10}{28 - 10 - 10} \times 10 = 45$ min (2).	1+1+1+2
27	Slant height of the cone $l = \sqrt{63^2 + 84^2} = 105$ cm (2); TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$ (2); TSA $= \frac{22}{7} \times 63 \times 105 + 2 \times \frac{22}{7} \times 63^2 = 45738$ cm^2 (1).	2+2+1
	OR: Median class is 20–30 with $f = 18$, and the cumulative frequency before it is $cf = 20 + x$ (2); $25 = 20 + \frac{50-(20+x)}{18} \times 10$ gives $x = 21$ (2); $x = 21$; then $y = 100 - 75 - x = 4$ (1).	2+2+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 24 cm	$H = 3h = 3 \times 8$; $H = 24$ cm
2	**(a)** 242	$l = 240$, $f_1 = 24$, $f_0 = 22$, $f_2 = 6$, $h = 20$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 240 + \frac{2}{20} \times 20 = 242$
3	**(d)** 5	Cross-multiplying: $25(11 + x) = 16(20 + x)$; $x = 5$
4	**(c)** 20 cm	Cancel $\frac{1}{3}\pi r^2$: $h = 2r$; $h = 20$ cm
5	**(c)** $20 + \frac{10}{18} \times 10$	$\frac{f_1 - f_0}{2f_1 - f_0 - f_2} = \frac{10}{32 - 14}$
6	**(d)** $\frac{20}{27}$	Favourable outcomes = number of red balls; $P = \frac{20}{27}$
7	**(b)** 9 : 25	$3^2 : 5^2 = 9 : 25$; Ratio $= 9 : 25$
8	**(b)** 25	$\frac{7890 + 110p}{75 + p} = 106.4$ gives $(110 - 106.4)p = 90$; $p = 25$
9	**(b)** $\frac{37}{80}$	Favourable numbers: 37 outcomes; number $= 37$; $P = \frac{37}{80}$
10	**(a)** 27 : 64	$3^3 : 4^3 = 27 : 64$; Ratio $= 27 : 64$
11	**(b)** 22	$70 = 60 + \frac{50 - (20 + x)}{16} \times 20$ gives $x = 22$; $x = 22$; then $y = 100 - 69 - x = 9$
12	**(d)** $\frac{17}{20}$	Favourable outcomes = number of good pens; $P = \frac{153}{180} = \frac{17}{20}$
13	**(d)** A is false but R is true.	Volume is conserved, not surface area (e.g. 8 small spheres have twice the surface area of the big one): A false; R true.
14	**(d)** A is false but R is true.	The modal class has the *highest* frequency, so A is false; R is true.
15	**(d)** A is false but R is true.	A is false since $P(E) \leq 1$; R states exactly that and is true.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\sum f_i = 98 + p$ and $\sum f_i x_i = 7730 + 75p$; $\frac{7730 + 75p}{98 + p} = 78.8$ gives $(75 - 78.8)p = -7.6$ (1); $p = 2$ (1).	1+1
17	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: 15 pairs; number $= 15$ (1); $P = \frac{15}{36} = \frac{5}{12}$ (1).	1+1
18	Volume of material = outer hemisphere − inner hemisphere; $\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(14^3 - 3.5^3)$ (1); Volume $= 5659.5$ cm^3 (1).	1+1
19	Highest frequency is 22, so the modal class is 220–240; $l = 220$, $f_1 = 22$, $f_0 = 13$, $f_2 = 7$, $h = 20$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 220 + \frac{9}{24} \times 20 = 227.5$ (1).	1+1
20	$P(\text{blue}) = \frac{x}{16 + x}$ and $P(\text{red}) = \frac{16}{16 + x}$; $\frac{x}{16 + x} = 5 \times \frac{16}{16 + x}$ gives $x = 5 \times 16$ (1); $x = 80$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$\frac{13 + x}{24 + x} = \frac{16}{27}$ (1); Cross-multiplying: $27(13 + x) = 16(24 + x)$ (1); $x = 3$ (1).	1+1+1
22	Volume of cylinder $= n\times$ volume of one cone (1); $\pi(12)^2(14) = n \times \frac{1}{3}\pi(6)^2(21)$ (1); $n = 8$ (1).	1+1+1
23	Median class is 50–60 with $f = 24$, and the cumulative frequency before it is $cf = 17 + x$ (1); $55 = 50 + \frac{50 - (17 + x)}{24} \times 10$ gives $x = 21$ (1); $x = 21$; then $y = 100 - 56 - x = 23$ (1).	1+1+1
24	Total possible outcomes $= 66$ (1); Favourable numbers: 47 outcomes; number $= 47$ (1); $P = \frac{47}{66}$ (1).	1+1+1
	OR: Volume of cylinder $= \pi \times 12^2 \times 24$ (1); Volume of a sphere $= \frac{4}{3}\pi(6)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 12$ (1).	1+1+1
25	$\sum f_i = 100$ and $\sum f_i x_i = 18680$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{18680}{100} = 186.8$ (1); $l = 180$, $f_1 = 22$, $f_0 = 21$, $f_2 = 21$, $h = 20$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 180 + \frac{1}{2} \times 20 = 190$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,6), (2,6), (3,6), (4,6), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6); number $= 11$ (1); $P = \frac{11}{36}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	(i) $4+5+6=15$ (1). (ii) $P(\text{red})=\frac{4}{15}$ (1). (iii) $P(\text{not green})=\frac{9}{15}=\frac{3}{5}$ (1). (iv) $P(\text{blue or green})=\frac{11}{15}$ (1); new total $=20$, red $=9$, $P=\frac{9}{20}$ (1).	1+1+1+2
27	$\sum f_i=100$ and $\sum f_i x_i=5530$; Mean $=\frac{\sum f_i x_i}{\sum f_i}=\frac{5530}{100}=55.3$ (2); $l=60$, $f_1=23$, $f_0=16$, $f_2=10$, $h=10$ (2); Mode: $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=60+\frac{7}{20}\times 10=63.5$ (1).	2+2+1
	OR: $\frac{9+x}{30+x}=\frac{8}{15}$ (2); Cross-multiplying: $15(9+x)=8(30+x)$ (2); $x=15$ (1).	2+2+1

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